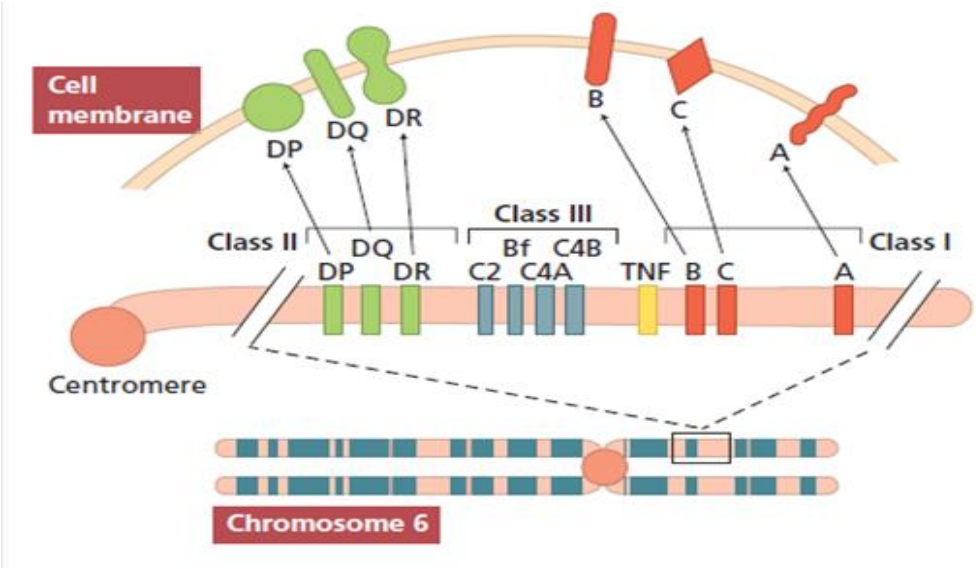
# MAJOR HISTOCOMPATIBILITY COMPLEX

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# History

It was known from the early days of **transplantation** that tissues, such as skin, exchanged between **non identical individuals** are **rejected**, whereas the **same grafts** between **identical twins** are **accepted**. This result showed that **tissue rejection** is a **genetically** determined process.




Prior to transplantation, laboratory tests, commonly **called HLA typing or tissue typing,** are performed to match the donor and the recipient.

**Human HLA complex**

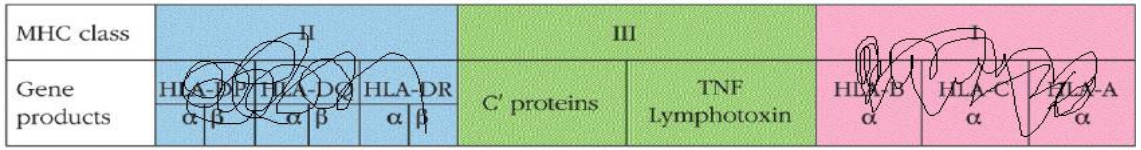
| MHC class | II | | | | | | III | | I | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Gene products | HLA-DP | | HLA-DQ | | HLA-DR | | C′ proteins | TNF Lymphotoxin | HLA-B | HLA-C | HLA-A |
| | α | β | α | β | α | β | | | α | α | α |

# Major histocompatibility complex (MHC)

- Originally known as 'histocompatibility antigens' because of the strong reactions they provoked during **mismatched** organ **transplantation**.
- Is also referred to as the **human leukocyte antigen (HLA) complex** in humans and as the **H2 complex** in mice.
- MHC molecules are cell-surface glycoproteins of two basic types: class I and class II.
- **MHC class III** molecules are a group of unrelated proteins that **do not** share structural **similarity** or **function** with **class I** and **II** molecules, although many of them do **participate** in other aspects of the **immune response**.

MHC class III locus encodes complement proteins and several inflammatory cytokines (eg. tumor necrosis factors).

**Human HLA complex**




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- MHC exhibit extensive genetic polymorphism with multiple alleles each locus → As a result, genetic variability between individuals is very great and most unrelated individuals possess different (HLA) molecules.
  - → **Organ transplantation**
  - → **pathogens**
  - → **Autoimmune diseases**

- It is rare for any two unrelated individuals to have identical sets of HLA alleles. This makes transplantation between individuals who are not identical twins quite challenging.
- Even with partial matches, physicians still need to administer heavy doses of immunosuppressive drugs to inhibit the strong rejection responses that typically follow tissue transplantation.

- Each person has **two HLA haplotypes** (i.e., two sets of these genes—one on the **paternal** and the other on the **maternal** chromosome 6).
- These genes are **highly polymorphic** (there are many alleles of the class I and class II genes).
- For example, as of 2020, there are at least 20,000 HLA Class I alleles and 7700 HLA Class II alleles, and more are being discovered. للاطلاع
- The **TCR** of an **individual T cell** will only **recognize** antigen as a complex of antigenic peptide and self-MHC.

The genes within the MHC locus exhibit a **co-dominant** form of expression, meaning that both maternal and paternal gene products (from both haplotypes) are expressed at the same time and in the same cells.

- Depending on the nature of the antigen (**endogenous** or **exogenous**), the MHC **response** is **different**.
- **Class I** molecules present on most **nucleated cells.** Key exceptions are observed on cells in the retina and brain.
- **Class II** molecules are expressed constitutively only by antigen-presenting cells (eg. macrophages, dendritic cells, and B cells); some other cell types can be induced to express class II molecules and to function as antigen-presenting cells under certain conditions.
- **Expression** of **MHC** molecules is **increased** or **decreased** (**regulated**) by cytokines produced during both innate and adaptive immune responses.
- **IFN-γ** increases the expression of both **MHC** molecules class I and II.
- **MHC expression is decreased** by **infection** with certain viruses, (in **cytomegalovirus infection**, for example, a viral protein binds to 2-microglobulin, preventing assembly of class I MHC molecules and their transport to the plasma membrane).

# The Phenomenon of MHC Restriction

- T cells must be **specific** not only for the **antigen** but also for **MHC** molecules, and T cell antigen recognition is restricted by the MHC molecules a T cell sees.
- Subsequent studies established that the recognition of antigens by **CD8+ CTLs** is restricted by **class I MHC** molecules, and the responses of **CD4+ helper T lymphocytes** to antigens are restricted by **class II MHC** molecules.

CD8+ cytotoxic T cells is restricted by



class I MHC molecules



Endogenous Ags



eg. viral infected cells, cancer cells

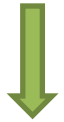
CD4+ helper T lymphocytes



class II MHC molecules



Exogenous Ags



eg. Extracellular pathogens

- The **MHC class I** antigens are divided into three groups (**A, B, and C**), and each group is controlled by a different gene locus within the MHC region on **chromosome 6** in human.

***HLA-A, HLA-B,* and *HLA-C***

**Human HLA complex**




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- All MHC class I antigens are made up of an α heavy chain, controlled by a gene in the relevant MHC locus, associated with a smaller chain called β2-microglobulin, controlled by a gene on chromosome 15.

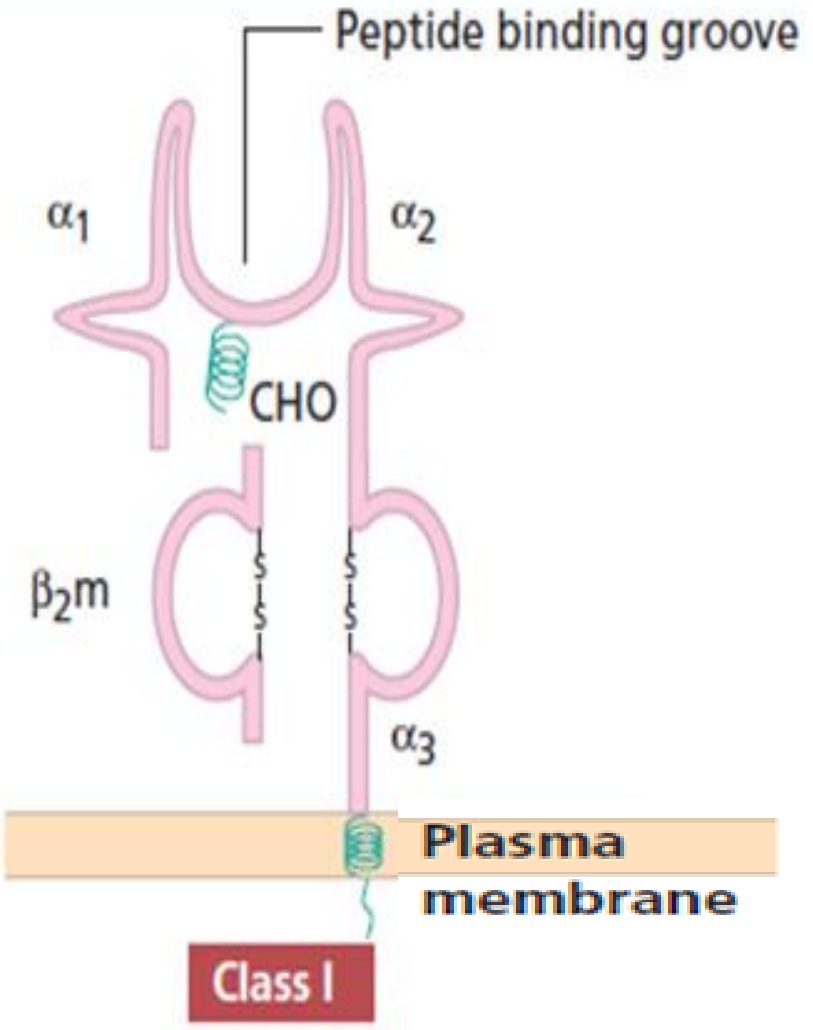
- The differences between individual MHC class I antigens are due to variations in the α chains; the β2- microglobulin component is constant.

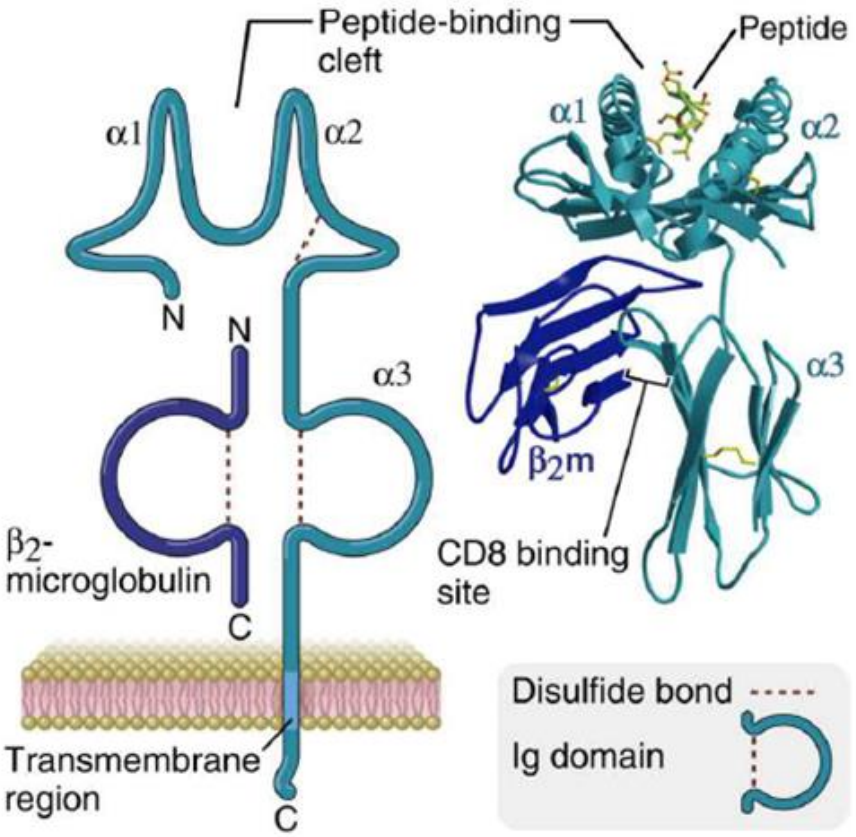
- The affinity (tightness of binding) of individual peptide binding depends on the nature and shape of the groove, and accounts for the MHC restriction mentioned earlier.

- Most individuals are heterozygous for MHC genes and therefore express six different class I MHC molecules on every cell, containing α chains encoded by the two inherited alleles of *HLA-A, B,* and *C* genes.

Diagrammatic representation of MHC class I

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# Antigen Processing and Presentation

- **Endogenous antigens** (including **viral antigens** that have infected host cells) are processed by the endoplasmic reticulum (ER) and presented by **MHC class I-bearing cells** exclusively to **CD8+ T cells**.

- Prior to presentation on the cell surface, **endogenous antigens** are broken down into **short peptides**, which are then actively transported from the cytoplasm to **ER** by proteins.

- These proteins act as a **shuttle** and are so named 'transporters associated with antigen processing' (TAP-1 and TAP-2).

- TAP proteins deliver peptides to **MHC class I** molecules in the end ER, from whence the complex of MHC and peptide is delivered to the **cell surface.**

***Mutations** that affect function in either **TAP gene** prevent surface expression of MHC class I molecules.

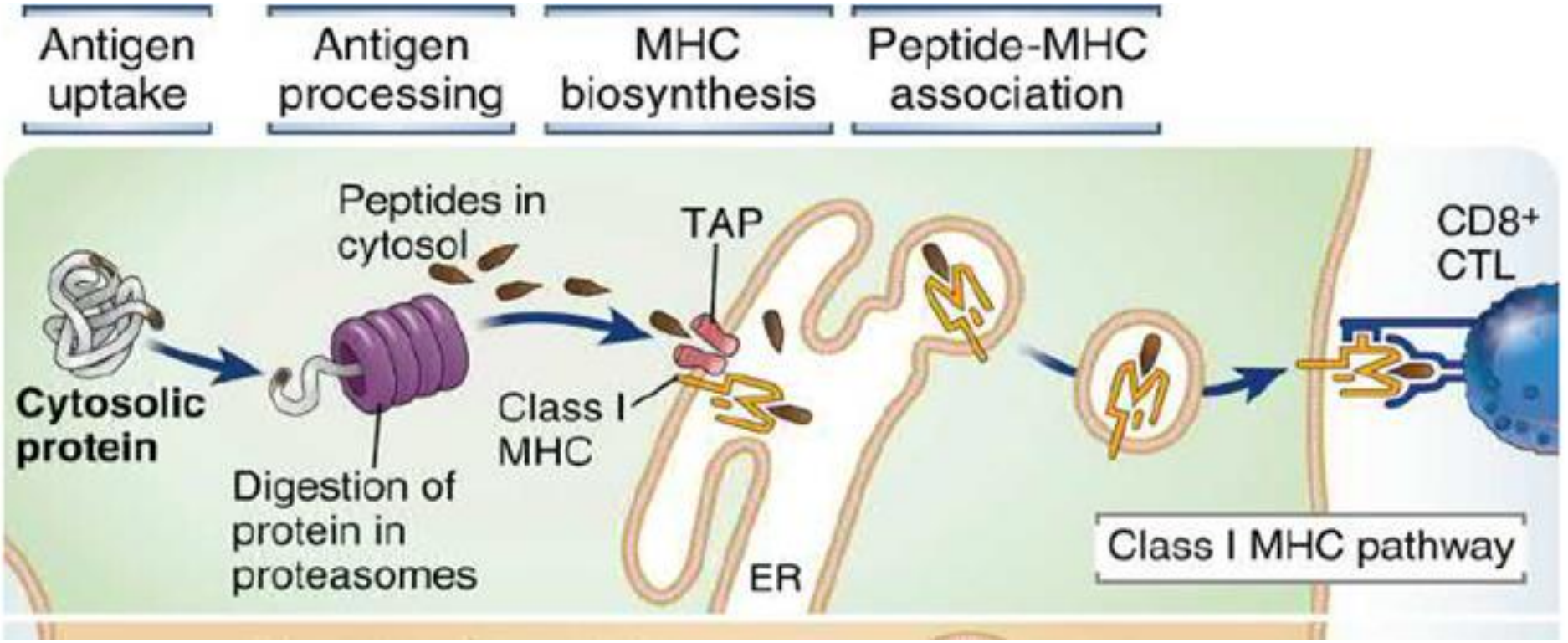
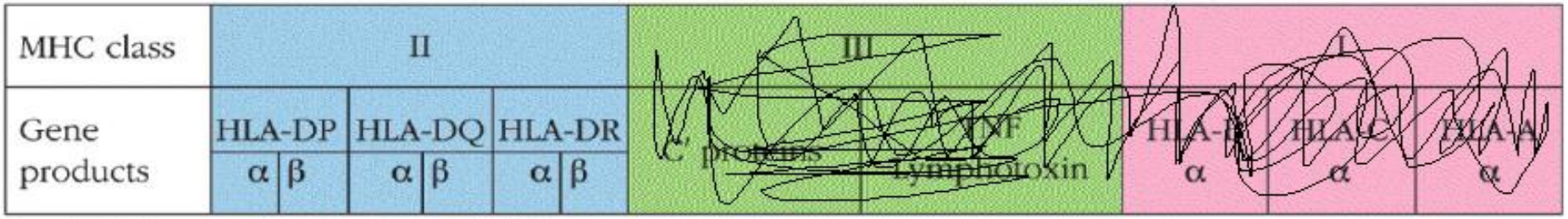



**FIGURE 6.13 Pathway of endogenous antigen processing and presentation.**

- **MHC class II molecules** are held in the ER.
- **MHC class II antigens** have two heavy chains, **α** and **β**, both coded for by genes in the MHC region of **chromosome 6**.
- In humans, there are three groups of class II antigens: namely, HLADP, HLA-DQ, and HLA-DR.
- Humans **inherit**, from each parent, one *DPA* and one *DPB* gene encoding, respectively, the α and β chains of an HLA-DP molecule (same feature is for DQ and DR)
- Thus, each **heterozygous** individual expresses six to eight pairs of class II MHC α and β chain molecules, one set each of *DP* and *DQ,* and one or two of *DR*.

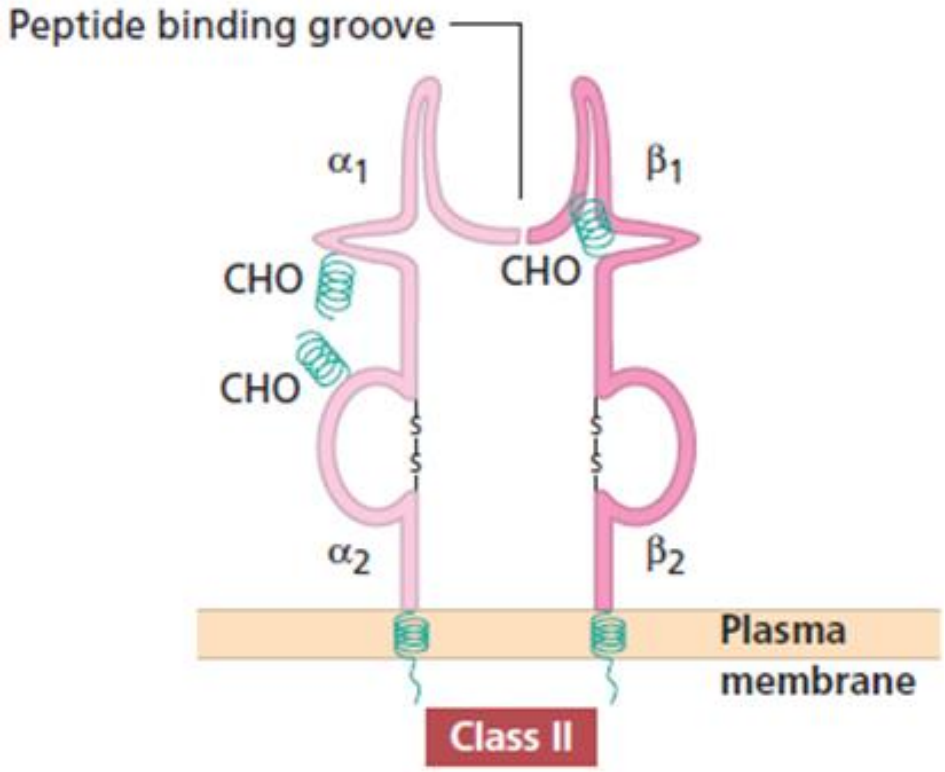
**Human HLA complex**

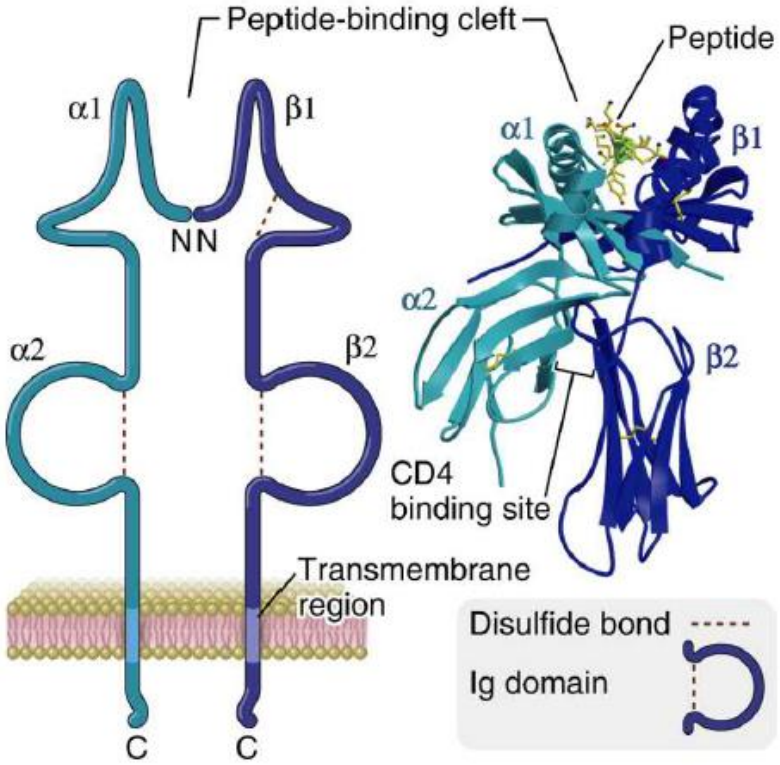



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Diagrammatic representation of MHC class II antigens.

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- **Exogenous antigens (**eg. bacteria**)** are processed by the **lysosomal** route and presented by **MHC class II** antigens to **CD4+ T cells.**

- **Newly synthesized** MHC class II molecules are held in the ER until they are ready to be transported to the cell surface.

- Whilst in the ER, class II molecules are **prevented** from binding to peptides in the lumen by a protein known as MHC class II-associated **invariant chain**.

- **The invariant chain** also directs delivery of class II molecules to the endosomal compartment where exogenous antigens are processed and made available for binding to class II molecules.

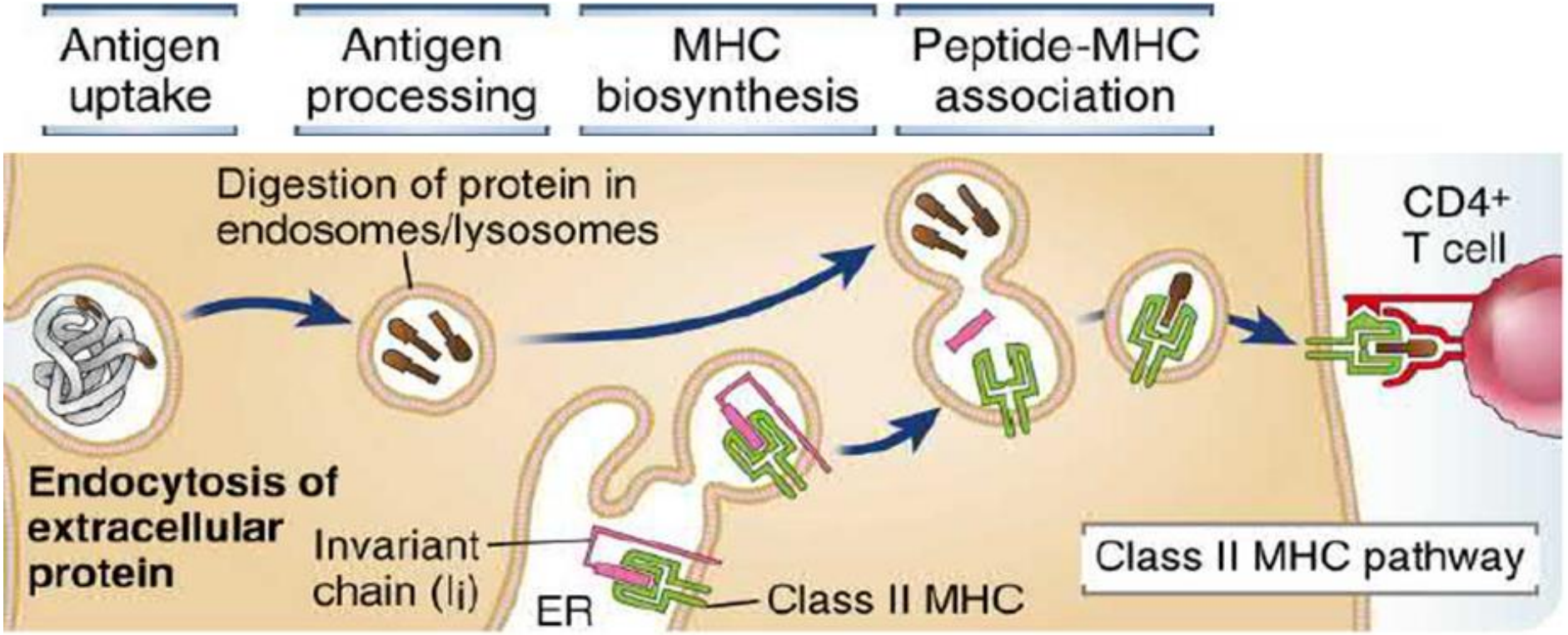



**FIGURE 6.13 Pathway of exogenous antigen processing and presentation.**

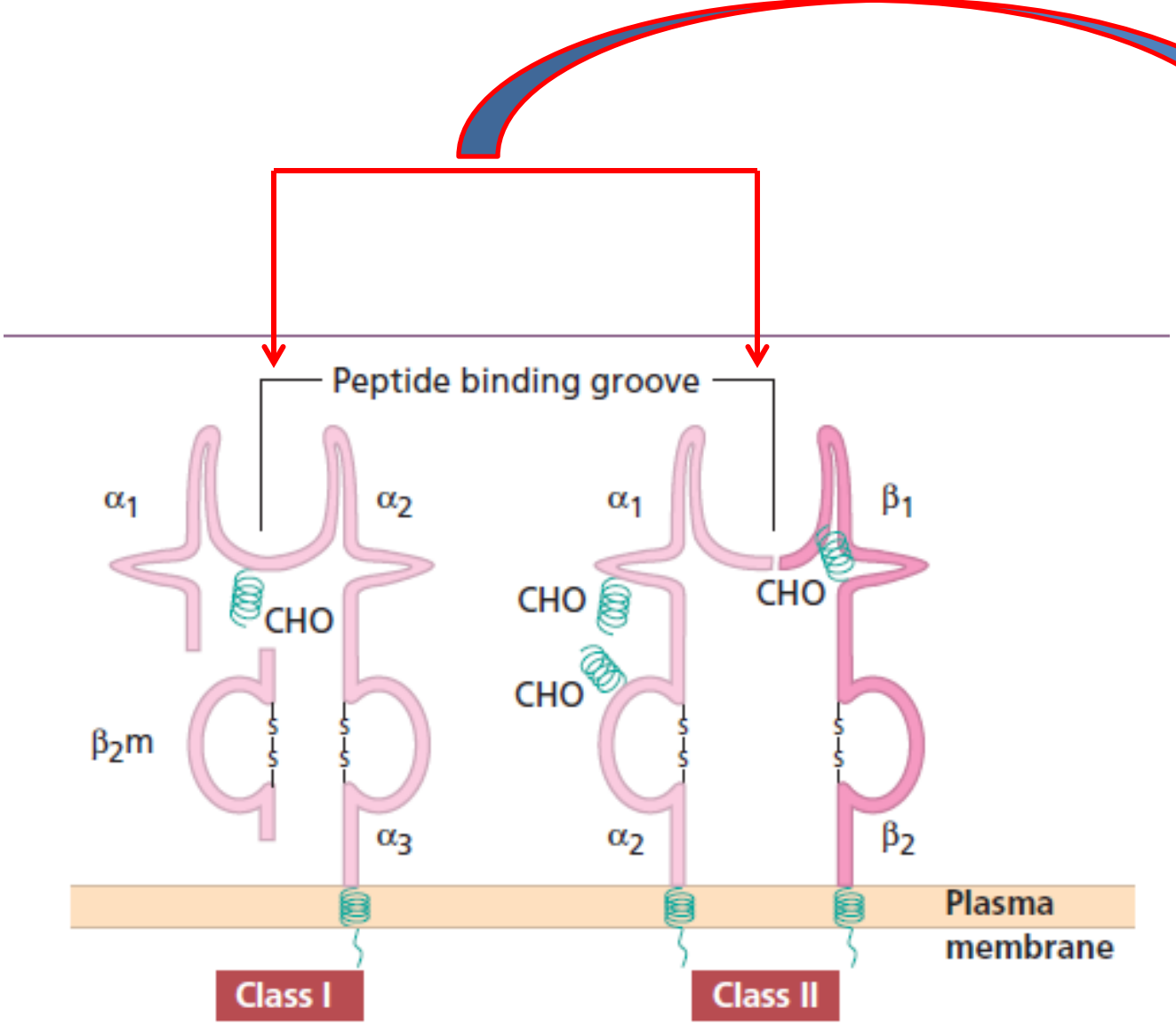



**Fig. 1.5** Diagrammatic representation of MHC class I and class II antigens. $\beta_2$m, $\beta_2$-microglobulin; CHO, carbohydrate side chain.

**Each class I and class II MHC molecule has a single peptide binding cleft that binds one peptide at a time, but each MHC molecule can bind many different peptides.**

- Some bacterial and viral antigens are able to activate large numbers of T cells through a special pathway.
- These proteins are called **superantigens**.
- **Superantigens** do not require processing and therefore are able to bind to MHC molecules **outside the peptide-binding cleft**.
- **Classic examples of superantigens include** certain bacterial toxins, including the staphylococcal enterotoxins, toxic shock syndrome toxin, and group A streptococcal pyrogenic exotoxin A.
- A consequence of this **massive activation of T cells** is the **overproduction of cytokines** which may contribute to a "**cytokine storm**" causing severe symptoms of shock and multiple organ failure.

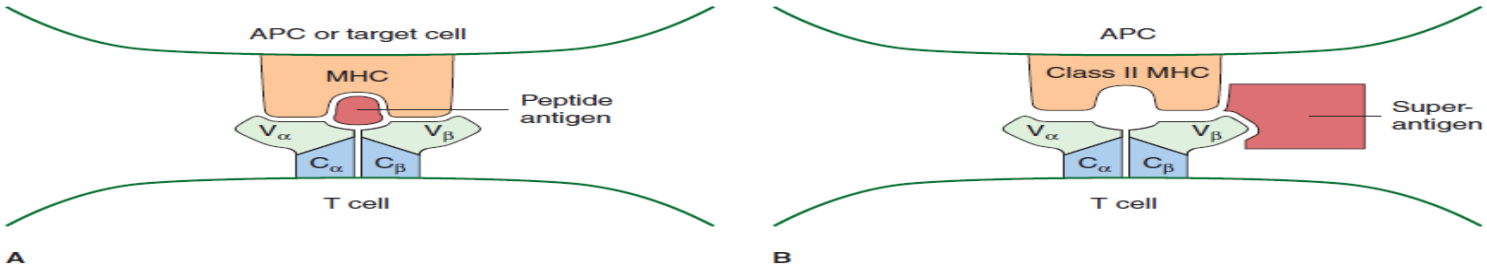



**FIGURE 8-4** Binding of antigen by MHC and T-cell receptor (TCR). In ***panel A***, a model of the interaction between peptide antigen, MHC, and the TCR is shown. The $V_\alpha$ and $V_\beta$ regions of the TCR are shown interacting with the α helices that form the peptide-binding groove of MHC. In ***panel B***, a model of the interaction between a superantigen, MHC, and the TCR, is shown. The superantigen interacts with the $V_\beta$ region of the TCR and with class II MHC outside the peptide-binding groove. (Adapted with permission from Stites DG, Terr AI, Parslow TG [editors]: *Medical Immunology*, 9th ed. McGraw-Hill, 1997, p. 132. © McGraw-Hill Education.)

# MHC and Disease Susceptibility

- Some HLA alleles occur at a **much higher frequency** in people suffering from certain **diseases** than in the general population.
- The diseases associated with particular MHC alleles include **autoimmune disorders**, certain **viral diseases**, some **neurologic disorders**, and several different **allergies**….but multiple genetic and environmental factors have roles in the development of disease, especially autoimmune diseases, with the MHC playing an important but not exclusive role.

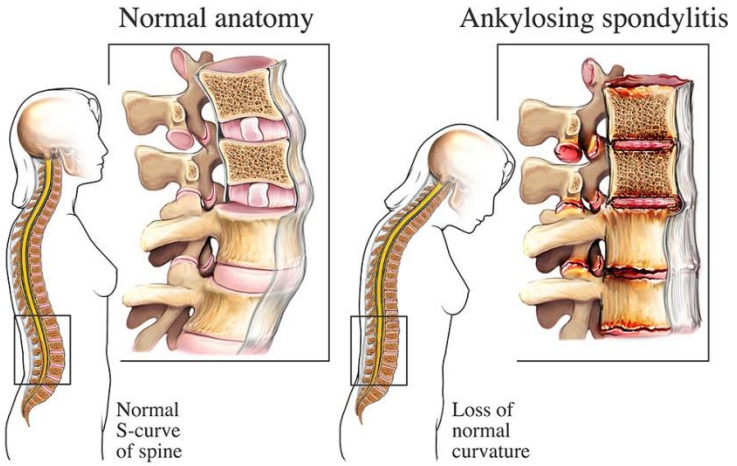
**For example:-**

- Rheumatoid arthritis occurs **predominantly** in individuals carrying the *HLA-DR4* gene.





- Ankylosing spondylitis is **more likely** to occur in people who carry HLA-B27.




- Asthma has been associated with HLA-DQB1 gene.



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# Comparison between MHC I and II

| Feature | Class I MHC Proteins | Class II MHC Proteins |
|---|---|---|
| Present antigen to CD4-positive cells | No | Yes |
| Present antigen to CD8-positive cells | Yes | No |
| Found on surface of all nucleated cells | Yes | No |
| Found on surface of "professional" antigen-presenting cells, such as dendritic cells, macrophages, and B cells | Yes1 | Yes |
| Encoded by genes in the HLA locus | Yes | Yes |
| Expression of genes is codominant | Yes | Yes |
| Multiple alleles at each gene locus | Yes | Yes |
| Composed of two peptides encoded in the HLA locus | No | Yes |
| Composed of one peptide encoded in the HLA locus and a $\beta_2$-microglobulin | Yes | No |

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